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BRIEF REPORT

Concurrent Agreement and Test–Retest Reliability of a Global-Positioning-System Device for Measuring Maximal Horizontal Deceleration Ability in Elite Youth Academy Soccer Players

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Purpose: Investigate the concurrent agreement and test–retest reliability of 10-Hz global-positioning-system (GPS) device against a criterion measure (47-Hz radar device) to assess maximal horizontal deceleration ability (maximum deceleration [DEC_{Max}], average deceleration [DEC_{Ave}], time to stop, and distance to stop). ***Methods:*** Thirty-two male elite youth academy soccer players (age 18.1 [1.6] y, body mass 76.6 [7.9] kg) completed the acceleration–deceleration ability test with 16 completing a second test to assess test–retest reliability. Maximal horizontal deceleration ability was measured concurrently using GPS_{Raw} (10-Hz data), GPS_{Export} (STATSports software), and a radar device. Bland–Altman method and equivalence testing assessed concurrent agreement and intraclass correlations with coefficient of variation (%) was used to assess test–retest reliability. ***Results:*** Equivalence testing showed mean difference between the radar device and GPS-derived values of DEC_{Ave} and DEC_{Max} were within equivalence bounds. GPS_{Raw} and GPS_{Export} derived values of DEC_{Max} showed *good* overall (intraclass correlations = .84–.86, coefficient of variation % = 4.50–5.48) test–retest reliability. ***Conclusion:*** Practitioners can consider using deceleration variables (DEC_{Ave} and DEC_{Max}) obtained from GPS as a cost-effective, valid, and reliable alternative to radar technology to assess maximal horizontal deceleration ability in team-sport players.

Keywords: braking, profiling, acceleration–deceleration ability test, football, validity

Profiling horizontal deceleration ability can inform performance-enhancement and injury-prevention strategies in team-sport environments.[1] However, there remains a paucity of valid and reliable methods to assess an athlete's horizontal deceleration ability in the field.[1] While radar or laser devices are considered the criterion measure of horizontal deceleration ability,[1] not all high-level clubs have access to such technologies.[2] Moreover, these technologies only permit individual testing, making it difficult for practitioners to assess horizontal deceleration ability within time-constrained environments.[3] Instead, most clubs are now equipped with global positioning system (GPS) devices,[2] which presents a viable alternative to assess horizontal deceleration capabilities without additional equipment and associated time demand.

Previous research has highlighted several variables to quantify horizontal deceleration ability.[1] The deceleration variables: maximum deceleration (DEC_{Max}), average deceleration (DEC_{Ave}), and time to stop (TTS) have all shown *moderate* intertest reliability using radar technology during a maximal deceleration task.[1] However, the reliability for many of these variables obtained from GPS remains unclear. *Poor* to *moderate* interunit reliability for DEC_{Max} has been reported previously, with these mixed results likely due to differences in protocols used.[4,5] Furthermore, the concurrent agreement and test–retest reliability of these deceleration variables obtained from a GPS device during a maximal deceleration task remains unknown.

Therefore, the aims of this study were to examine (1) the concurrent agreement between deceleration variables obtained from GPS with a criterion measure and (2) the test–retest reliability of deceleration variables assessed using a GPS device in elite youth academy soccer players.

Methods

Subjects

Thirty-two male youth soccer players (age: 18.1 [1.6] y; body mass: 76.6 [7.9] kg) from an English Premier League academy were recruited and completed the concurrent agreement session. Due to congested fixtures, only 16 players (age: 17.4 [1.3] y; body mass: 73.6 [8.0] kg) completed the test–retest reliability session. The research was approved by St Mary's University ethics committee and conducted in accordance with the Declaration of Helsinki. All participants provided voluntary informed consent prior to starting the study.

Design

We used a within-subject repeated-measures design.

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Methodology

The study was conducted during an in-season competition phase, with testing completed at the same time of day on an artificial turf surface. Participants all wore studded footwear and completed a 20-m maximal sprint test followed by 2 testing sessions, each separated by 1 week (Figure 1). Before testing, participants completed a 10-minute standardized warm-up and 2 progressive deceleration test trials.

Maximal 20-m sprint times were recorded using timing gates (TC, Brower Timing Systems) positioned at 0 and 20 m. Participants started 1 m behind the first gate and initiated their sprint from a stationary split stance, completing 2 trials with at least 2-minute recovery. The fastest 20-m split was used as a "criterion" time in the maximal horizontal deceleration test.[2]

During both testing sessions, participants completed 2 repetitions of the acceleration–deceleration ability test following a similar protocol used by Harper et al[1] (Figure 1). Trials were considered unsuccessful if the 20-m time was 5% greater than the "criterion" time and repeated after a 3-minute recovery.

Raw velocity data were recorded concurrently during the acceleration–deceleration ability test using 2 methods: A radar device (Stalker Pro II, Applied Concepts Inc) positioned 5 m behind the participants on a tripod 1 m off the ground (Figure 1), sampled data at 46.875 Hz and a 10-Hz GPS unit (APEX, STATSports) worn in a fitted vest. The average horizontal dilution of precision was 0.55 (0.38) and number of satellites was 20.3 (1.4).

Raw velocity data from the radar and GPS (GPS_{Raw}) were exported and processed in R statistical software (R, version 3.3.0) using a custom-made R-script to calculate the deceleration variables (DEC_{Ave}, DEC_{Max}, TTS, and distance to stop [DTS]) based on methods in previous studies (see Supplementary Material [available online]).[1,6,7] Maximum velocity (V_{Max}) and DEC_{Max} were also directly obtained from STATSports software (Sonra 4.0, STATSports; GPS_{Export}).

Statistical Analysis

Concurrent agreement between the criterion (radar) and practical measures (10-Hz GPS_{Raw}/GPS_{Export}) was assessed with Bland–Altman method limits of agreement (LOA; 95%) and equivalence testing (two one-sided tests).[8] Two one-sided tests lower and upper equivalent bounds were based on the smallest effect size in raw units and test–retest typical error of the criterion measure[1]: $V_{Max} = \pm 0.15$ m·s^{-1}, TTS = ±0.1 s, DTS = ±0.5 m, $DEC_{Ave} = \pm 0.25$ m·s^{-2}, and $DEC_{Max} = \pm 0.50$ m·s^{-2}. A secondary agreement analysis was also conducted (see Supplementary Material [available online]).

Test–retest reliability of the practical measures (10-Hz GPS_{Raw}/GPS_{Export}) was assessed by determining the intraclass correlation coefficient (ICC; 2-way mixed model), coefficient of variation (CV%), and the standard error of measurement. CV% was calculated from the typical error, expressed as a percentage,[1,2] and standard error of measurement was calculated using the formula: $SD \times \sqrt{(1 - ICC)}$, expressed in raw units. Overall reliability was interpreted as follows: ICC > .9 and CV% < 5 = excellent; ICC .75 to .9 and CV% < 10 = good; ICC < .75 or CV% < 10 = moderate; ICC < .75 and CV% > 10 = poor.[9,10]

For information regarding the calculation of agreement and test–retest reliability measures, see Supplementary Material (available online).

Results

Data related to the concurrent agreement and test–retest reliability are displayed in Table 1. LOA graphs from the Bland–Altman analysis are shown in Figure 2.

Discussion

The main findings of this study suggest GPS as a potential valid and reliable device to assess DEC_{Ave} and DEC_{Max} variables. Therefore, opening-up the possibilities for practitioners assess maximal horizontal deceleration ability in the field.

Concurrent agreement and equivalence analysis showed the mean difference between the radar and GPS for DEC_{Ave} and DEC_{Max} were within the equivalence bounds, suggesting devices to be practically equivalent. However, the LOAs were outside these bounds, indicating a potential lack of between-method agreement on an individual level. Crang et al[11] found similar mean difference (–0.07 m·s^{-2}) between a 10-Hz GPS and laser device measuring DEC_{Max}, supporting the current results. However, the decelerations were not performed from high speeds (<7 m·s^{-1}),[11] which may explain the greater mean difference found in the current study. This suggests the ability of GPS devices to adequately detect deceleration may be compromised when movement velocity is increased.[3,12] In support of this, higher sampling devices (eg, 16 Hz) still exhibited error (typical error of estimate = 1.59% [0.42%]) in V_{Max} during high-velocity movements.[3] Therefore, practitioners may still wish to use GPS-derived values of DEC_{Max} and DEC_{Ave} to assess an athlete's horizontal deceleration ability. However, *poor* agreement (mean difference and LOA outside of equivalence bounds) was found between the radar and GPS for DTS and TTS. The origin of this difference is unclear but may be attributed to the higher sampling rate (47 Hz vs 10 Hz) of the radar compared to the GPS or data-processing methods. Therefore, their use in practice should be carefully considered.

The present study demonstrated *good* overall test–retest reliability for DEC_{Max}. In contrast, previous studies have shown *poor* to *moderate* interunit reliability in values of DEC_{Max}.[4,5] However, neither of these studies controlled the deceleration velocities (eg, participants achieving ≥ 95% of 20-m split time) therefore making it difficult to compare the current results. Nonetheless, the current findings suggest GPS_{Export} values of DEC_{Max} can be used as a time-efficient method (no additional data processing) that could be easily implemented during in situ pitch-based warm-ups to regularly monitor player horizontal deceleration ability. In addition, practitioners may wish to use GPS_{Raw} data to calculate novel variables such as DEC_{Ave}, TTS, and DTS. Although all these variables showed *moderate* test–retest reliability, similar reliability (CV% = 6.0) in DEC_{Ave} measured using a 10-Hz GPS unit has been observed previously,[12] potentially supporting its use in practice. However, using GPS_{Raw} data requires additional postprocessing and a custom R-script to calculate these variables which may limit accessibility practitioners. In addition, when using radar technology, Harper et al[1] found similar test–retest reliability values for all deceleration variables compared with those found in the current study. This suggests error from the testing procedure (eg, deceleration strategy) rather than the measurement technique itself.

Practical Applications

- DEC_{Ave} and DEC_{Max} variables obtained from GPS devices displayed equivalent mean difference compared with a radar

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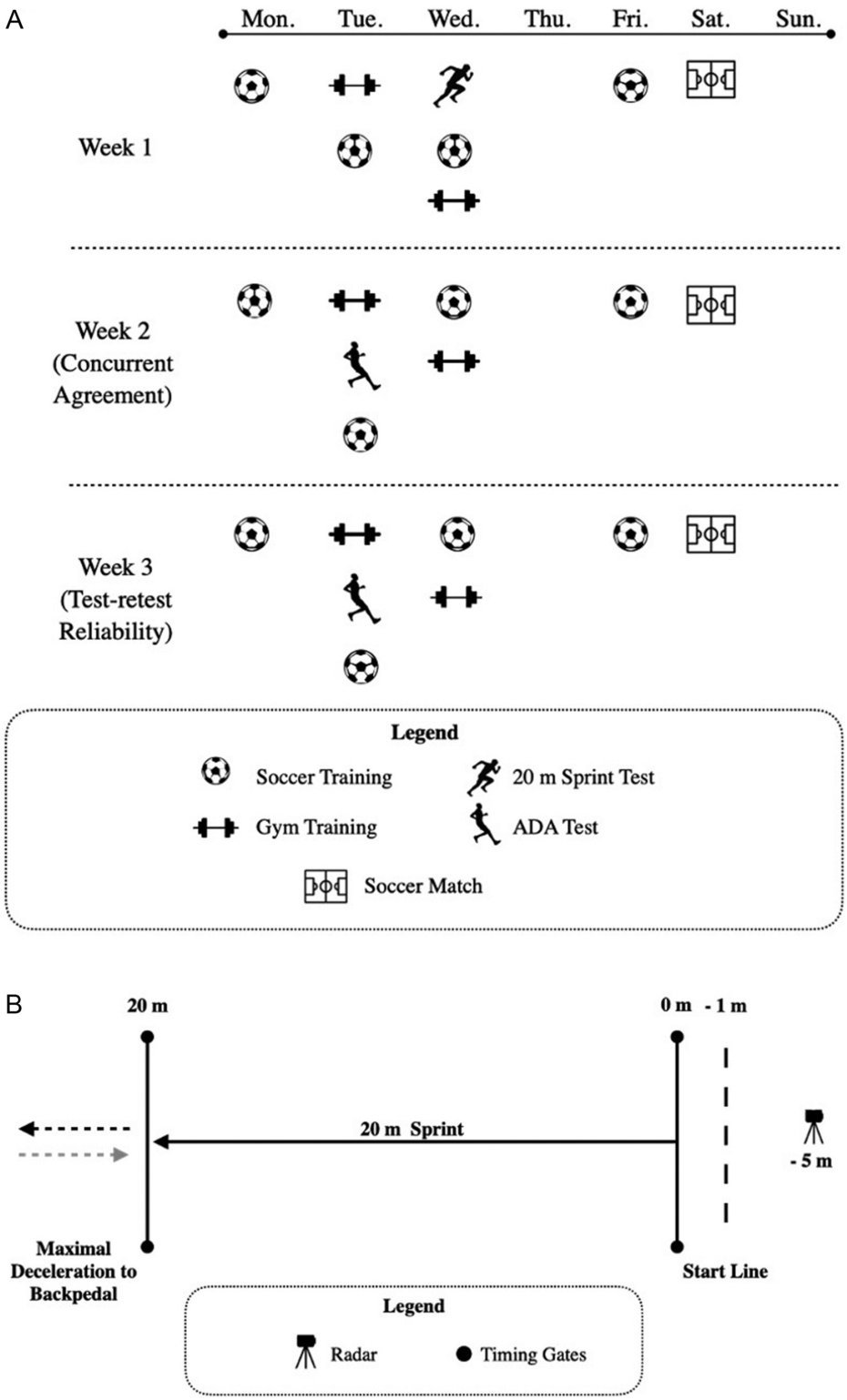

Figure 1 — (A) Schematic representation of the study design. Testing was conducted within a 3-week regular soccer training plan. Week 1 included a maximal 20-m linear-sprint test, and maximal horizontal deceleration ability was assessed using GPS (global-positioning-system) devices during weeks 2 and 3. (B) Schematic representation of the ADA test layout. Timing gates were positioned at 0 and 20 m. Participants were instructed to perform a maximal sprint over 20 m using the same starting procedure as the 20-m sprint test, followed by a maximal horizontal deceleration phase. The 20-m timing gate was used by the participants to initiate the deceleration phase. After the end of the deceleration, participants immediately backpedaled to the 20-m timing-gate line. ADA indicates acceleration–deceleration ability test.

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Table 1 Concurrent Agreement and Test–Retest Reliability Analysis

		Agreement vs radar				Test–retest reliability					
	Mean (SD)	Mean difference (95% CI)	TOST (lower and upper *P*)	Effect size (95% CI)	LOA (lower and upper bounds)	Day 1	Day 2	ICC (95% CI)	CV% (95% CI)	Rating	SEM (95% CI)
Radar (criterion)											
V_{Max}, m·s^{-1}	8.07 (0.26)	—		—	—	8.16 (0.21)	8.09 (0.25)	.88 (.69 to .96)	1.06 (0.78 to 1.65)	Good	0.08 (0.06 to 0.12)
TTS, s	1.43 (0.14)	—		—	—	1.40 (0.13)	1.34 (0.12)	.67 (.28 to .87)	5.45 (4.00 to 8.56)	Moderate	0.07 (0.05 to 0.11)
DTS, m	7.86 (0.73)	—		—	—	7.77 (0.65)	7.58 (0.63)	.62 (.20 to .85)	5.48 (4.02 to 8.61)	Moderate	0.39 (0.29 to 0.60)
DEC_{Ave}, m·s^{-2}	–4.33 (0.41)	—		—	—	–4.41 (0.41)	–4.39 (0.28)	.73 (.39 to .88)	4.36 (3.20 to 6.3)	Moderate	0.18 (0.13 to 0.28)
DEC_{Max}, m·s^{-2}	–7.58 (0.89)	—		—	—	–7.52 (0.97)	–7.64 (0.77)	.78 (.48 to .92)	6.04 (4.43 to 9.51)	Good	0.41 (0.30 to 0.63)
GPS_{Raw}											
V_{Max}, m·s^{-1}	7.80 (0.30)	0.28 (0.24 to 0.31)	Lower: $P < .001$ Upper: $P = 1.000$	1.87 (1.62 to 2.18)	0.28 (–0.01 to 0.56)	7.90 (0.25)	7.78 (0.23)	.81 (.54 to .93)	1.41 (1.04 to 2.18)	Good	0.10 (0.07 to 0.15)
TTS, s	1.61 (0.15)	–0.17 (–0.22 to –0.13)	Lower: $P = .999$ Upper: $P < .001$	–1.01 (–1.36 to –0.70)	0.33 (–0.51 to 0.16)	1.59 (0.14)	1.56 (0.09)	.48 (.00 to .78)	5.77 (4.23 to 9.07)	Moderate	0.09 (0.07 to 0.14)
DTS, m	8.25 (0.96)	–0.39 (–0.67 to –0.10)	Lower: $P = .209$ Upper: $P < .001$	–0.36 (–0.70 to –0.04)	2.11 (–2.49 to 1.72)	8.21 (0.85)	7.97 (0.57)	.53 (.06 to .81)	6.63 (4.85 to 10.44)	Moderate	0.49 (0.36 to 0.76)
DEC_{Ave}, m·s^{-2}	–4.29 (0.40)	–0.04 (–0.15 to 0.06)	Lower: $P < .001$ Upper: $P < .001$	–0.11 (–0.37 to 0.15)	0.79 (–0.83 to 0.75)	–4.37 (0.40)	–4.38 (0.35)	.57 (.13 to .83)	6.07 (4.45 to 9.56)	Moderate	0.25 (0.18 to 0.39)
DEC_{Max}, m·s^{-2}	–7.83 (1.01)	0.25 (0.08 to 0.41)	Lower: $P < .001$ Upper: $P = .002$	0.39 (0.22 to 0.57)	1.23 (–0.99 to 1.48)	–7.91 (1.06)	–8.05 (1.03)	.86 (.64 to .95)	5.48 (4.03 to 8.63)	Good	0.39 (0.29 to 0.60)
GPS_{Export}											
V_{Max}, m·s^{-1}	8.06 (0.33)	0.02 (–0.03 to 0.07)	Lower: $P < .001$ Upper: $P < .001$	0.08 (–0.07 to 0.24)	0.37 (–0.36 to 0.39)	8.16 (0.26)	8.03 (0.28)	.82 (.56 to .93)	1.51 (1.11 to 2.35)	Good	0.11 (0.08 to 0.17)
DEC_{Max}, m·s^{-2}	–7.46 (0.74)	–0.12 (–0.27 to 0.03)	Lower: $P < .001$ Upper: $P < .001$	–0.21 (–0.39 to –0.02)	1.13 (–1.25 to 1.01)	–7.51 (0.78)	–7.61 (0.73)	.84 (.59 to .94)	4.50 (3.31 to 7.05)	Good	0.30 (0.22 to 0.46)

Abbreviations: CV, coefficient of variation; DEC_{Ave}, average deceleration; DEC_{Max}, maximum deceleration; DTS, distance to stop; GPS, global positioning system; LOA, limits of agreement; ICC, intraclass correlation; SEM, standard error of measurement; TOST, two 1-sided tests; TTS, time to stop; V_{Max}, maximal velocity. Note: Raw data for criterion (radar) and practical (GPS_{Raw} and GPS_{Export}) are presented as mean (SD). Mean difference, effect size, LOA, and TOST equivalence testing are presented. Mean difference and effect size are presented with 95% CIs. TOST are presented as lower and upper *P* values; LOA presented as ±95% bounds. Reliability statistics are presented with 95% CIs.

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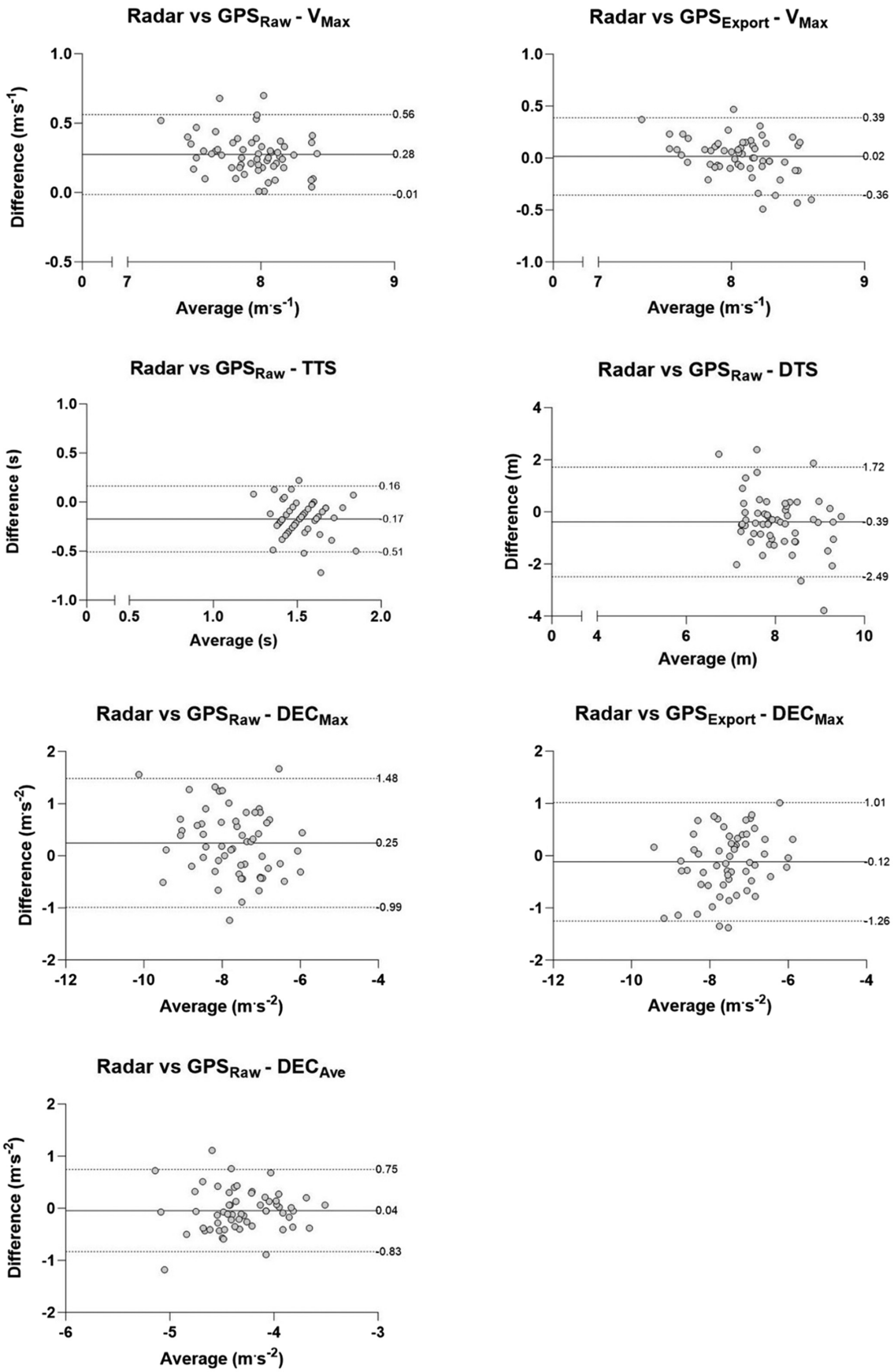

Figure 2 — Bland–Altman analyses. Black lines represent the mean differences; dashed lines, 95% limits of agreement; DEC_{Ave}, average deceleration; DEC_{Max}, maximum deceleration; DTS, distance to stop; GPS, global positioning system; TTS, time to stop; V_{Max}, maximal velocity.

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device. Therefore, suggesting a cost-effective alternative to radar technology for assessing horizontal deceleration ability.

- DEC_{Max} obtained with GPS showed *good* overall test–retest reliability, confirming GPS can be used to accurately monitor maximal horizontal deceleration ability over time,

while GPS devices were shown to be reliable in measuring DEC_{Max} from a 20-m sprint distance. Future investigations are needed to confirm the practical application of GPS from other sprint distances. Additionally, as only test–retest reliability was assessed, further research is necessary to understand the sensitivity to changes (eg, pre–post preseason) of the deceleration variables obtained with GPS.

Conclusion

The present study indicates that GPS-derived values of average and maximal deceleration as a potential cost-effective alternative to radar technology to assess maximal horizontal deceleration ability in elite youth soccer players. Future studies are needed to examine the use of GPS devices in measuring deceleration variables from different sprint distances to help inform more advanced insights into athletes' deceleration capabilities.

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